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Cryptology Assignment 3

1. Determine the multiplicative inverse modulo 26 for each of the following integers

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 mod 26

Ans:

(i) Multiplicative inverse of 1

$$26 = 1(26) + 0$$

$$1(1) = 1$$

$$1^{-1} \equiv 1 \pmod{26}$$

Multiplicative inverse of 1 = 1

(ii) Multiplicative inverse of 3

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2 \quad - \textcircled{1}$$

$$2 = 26 - 3(8)$$

$$1 = 3 - (26 - 3(8))$$

$$1 = 3 - 26 + 3(8)$$

$$1 = 3(9) + 26(-1)$$

$$3^{-1} \equiv 9 \pmod{26}$$

Multiplicative inverse of 3 = 9

(iii) Multiplicative inverse of 5

$$26 = 5(5) + 1$$

$$1 = 26 - 5(5)$$

$$1 = 5(5) + 26(-1)$$

$$-5 + 26 = 21$$

$$5^{-1} \equiv 21 \pmod{26}$$

Multiplicative inverse of 5 = 21

(iv) Multiplicative inverse of 7,

$$26 = 7(3) + 5, \quad 5 = 26 - 7(3)$$

$$7 = 5(1) + 2, \quad 2 = 7 - 5 \quad \text{--- ②}$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2) \quad \text{--- ①}$$

$$2 = 7 - 5 \quad \text{--- sub ② in ①}$$

$$1 = 5 - 2 \times 7 + 2 \times 5$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = 3 \times 26 - 3 \times (7(3)) - 2(7)$$

$$1 = 3 \times 26 - 11 \times 7$$

$$1 = (-11) \times 7 + 3 \times 26, \quad -11 + 26 = 15$$

$$7^{-1} \equiv 15 \pmod{26}$$

Multiplicative inverse of 7 is 15

(v) Multiplicative inverse of 9,

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8, \quad 8 = 26 - 9(2)$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 9 + 9(2) - 26$$

$$1 = 3(9) - 26$$

$$9^{-1} \equiv 3 \pmod{26}$$

Multiplicative inverse of 9 is 3

(vi) Multiplicative inverse of 11,

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3, \quad 3 = 11 - 4(2)$$

$$4 = 3(1) + 1, \quad 4 = 26 - 11(2)$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2)) = 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = 3 \times 26 - 3 \times (11(2)) - 11$$

$$1 = 3 \times 26 - 7 \times 11$$

$$1 = (-7)11 + 3 \times 26, \quad -7 + 26 = 19$$

$$11^{-1} \equiv 19 \pmod{26}$$

Multiplicative inverse of 11 is 19

(vii) Inverse of 15,

$$26 = 15(1) + 11, \quad 11 = 26 - 15$$

$$15 = 11(1) + 4, \quad 4 = 15 - 11(1)$$

$$11 = 4(2) + 3, \quad 3 = 11 - 4(2)$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2))$$

$$1 = 3(4) - 11$$

$$1 = 3(15 - 11(1)) - 11$$

$$1 = 3 \times 15 - 4 \times 11$$

$$1 = 3 \times 15 - 4(26 - 15)$$

$$1 = 7(15) - 4(26)$$

$$15^{-1} \equiv 7 \pmod{26}$$

Multiplicative inverse of 15 is 7

(viii) Inverse of 17,

$$26 = 17(1) + 9, \quad 9 = 26 - 17(1)$$

$$17 = 9(1) + 8, \quad 8 = 17 - 9(1)$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8$$

$$1 = 9 - (17 - 9(1))$$

$$1 = 2(9) - 17$$

$$1 = 2(26 - 17(1)) - 17$$

$$1 = 2 \times 26 - 2 \times (17(1)) - 17$$

$$1 = 2 \times 26 - 23 \times (17)$$

$$26 - 3 \times 17 = 1 = (-3)17 - 2(26), \quad 17^{-1} \equiv 23 \pmod{26}$$

Multiplicative inverse of 17 is 23

(ix) Inverse of 19,

$$\begin{aligned}26 &= 19(1) + 7, & 7 &= 26 - 19 \\19 &= 7(2) + 5, & 5 &= 19 - 7(2) \\7 &= 5(1) + 2, & 2 &= 7 - 5(1) \\5 &= 2(2) + 1.\end{aligned}$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 5 - 2 \times 7 + 2(5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(19 - 7(2)) - 2(7)$$

$$1 = 3 \times 19 - 8(7)$$

$$1 = 3(19) - 8(26 - 19)$$

$$1 = 11(19) - 8(26)$$

$$19^{-1} \equiv 11 \pmod{26}$$

Multiplicative inverse of 19 is 11

(x) Inverse of 21,

$$26 = 21(1) + 5, \quad 5 = 26 - 21$$

$$21 = 5(4) + 1$$

$$1 = 21 - 5(4)$$

$$1 = 21 - 4(26 - 21)$$

$$1 = 21 - 4(26) + 4(21)$$

$$1 = 5(21) - 4(26)$$

$$21^{-1} \equiv 5 \pmod{26}$$

Multiplicative inverse of 21 is 5

(xi) Inverse of 23,

$$26 = 23(1) + 3, \quad 3 = 26 - 23$$

$$23 = 3(7) + 2, \quad 2 = 23 - 3(7)$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2(1)$$

$$1 = 3 - (23 - 3(7))$$

$$1 = 8(3) - 23$$

$$1 = 8(26 - 23) - 23$$

$$1 = (-9)23 + 26(8)$$

$$26 - 9 = 17$$

$$23^{-1} \equiv 17 \pmod{26}$$

Multiplicative Inverse of 23 is 17

(xii) Inverse of 25,

$$25 = 25(1) + 1$$

$$1 = 26 - 25$$

$$1 = (-1)25 + 26$$

$$26 - 1 = 25$$

$$25^{-1} \equiv 25 \pmod{26}$$

Inverse of 25 is 25.

2. Choose any string of your choice to encrypt using the affine cipher with key values multiplier 5 and Additive 8

Ans: For Encryption,

$$P = [CK_1 * P + K_2] \pmod{26}$$

Given $K_1 = 5$, $K_2 = 8$

Plaintext		$K_1 \times P$	$+ K_2$		$\pmod{26}$	
G	6	$5 \times 6 = 30$	$30 + 8 = 38$	38	12	m
O	14	$5 \times 14 = 70$	$70 + 8 = 78$	78	0	a
O	14	$5 \times 14 = 70$	$70 + 8 = 78$	78	0	a
D	3	$5 \times 3 = 15$	$15 + 8 = 23$	23	23	x
L	11	$5 \times 11 = 55$	$55 + 8 = 63$	63	11	L
U	20	$5 \times 20 = 100$	$100 + 8 = 108$	108	4	e
C	2	$5 \times 2 = 10$	$10 + 8 = 18$	18	18	s
K	10	$5 \times 10 = 50$	$50 + 8 = 58$	58	6	g

Ciphertext : M A X L E S G

3. Encrypt "HI" using the Affine Cipher with $a=11$ and $b=5$

Ans: For Encryption:

$$C = E(K_1 * P + K_2) \bmod 26$$

$$K_1 = 11, K_2 = 5$$

Plaintext		$K_1 \times P$		$+ K_2$		$\bmod 26$	Cipher
H	7	$11 \times 7 = 77$		$77 + 5 = 82$		4	E
I	8	$11 \times 8 = 88$		$88 + 5 = 93$		15	P

cipher text : EP

4. The ciphertext UCR was encrypted using the affine function $9x + 2 \bmod 26$. Find the plaintext.

Ans: For Decryption, $P = K^{-1} (C - K_2) \bmod 26$

$$K_1 = 9, K_2 = 2$$

$$9^{-1} \bmod 26$$

$$26 = 9(2) + 8, 8 = 26 - 9(2)$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 3(9) - 26$$

$$9^{-1} \bmod 26 = 3$$

Ciphertext		$K^{-1} (C - K_2)$		$\bmod 26$	Plaintext
U	20	$3(20 - 2) = 54$		2	C
C	2	$3(2 - 2) = 0$		0	a
R	17	$3(17 - 2) = 45$		19	T

Plaintext : CAT

5. Decrypt the cipher text using Affine cipher, K D Y X J U W D E X Z Y using the Key 3 (Mul) & 4

Ans For Decryption,

$$P = K^{-1}(CC - K_2) \bmod 26$$

Cipher text : K D Y X J U W D E X Z Y

$$K_1 = 3 \quad K_2 = 4$$

$$3^{-1} \bmod 26$$

$$26 = 3(8) + 2, \quad 2 = 26 - 3(8)$$

$$3 = 2(1) + 1$$

$$1 = 3 - 2$$

$$1 = 3 - (26 - 3(8))$$

$$1 = 3(9) - 26$$

$$3^{-1} \bmod 26 = 9$$

Cipher text		$K^{-1}(CC - K_2) \bmod 26$		Plaintext
K	10	$9(10 - 4) = 54$	2	C
D	3	$9(3 - 4) = -9$	17	R
Y	24	$9(24 - 4) = 180$	24	Y
X	23	$9(23 - 4) = 171$	15	P
J	9	$9(9 - 4) = 45$	19	T
U	20	$9(20 - 4) = 144$	14	O
W	22	$9(22 - 4) = 162$	6	G
D	3	$9(3 - 4) = -9$	17	R
E	4	$9(4 - 4) = 0$	0	A
X	23	$9(23 - 4) = 171$	15	P
Z	25	$9(25 - 4) = 189$	7	H
Y	24	$9(24 - 4) = 180$	24	Y

Plaintext : CRYPTOGRAPHY

6. Use $K_1 = 13$ (Multiplier), $K_2 = 4$ (Even though it is not a valid key - try to encrypt).

Encrypt the string "input" & "alter" using the above key & find the what is the issue.

Ans: For Encryption,

$$C = E(K_1 * P + K_2) \bmod 26$$

$$K_1 = 13, K_2 = 4$$

Plain Text		$K_1 * P$	$+ K_2$		
I	8	$13 \times 8 = 104$	$104 + 4 = 108$	4	E
N	13	$13 \times 13 = 169$	$169 + 4 = 173$	17	R
P	15	$13 \times 15 = 195$	$195 + 4 = 199$	17	R
U	20	$13 \times 20 = 260$	$260 + 4 = 264$	4	E
T	19	$13 \times 19 = 247$	$247 + 4 = 251$	17	R
A	0	$13 \times 0 = 0$	$0 + 4 = 4$	4	E
L	11	$13 \times 11 = 143$	$143 + 4 = 147$	17	R
T	19	$13 \times 19 = 247$	$247 + 4 = 251$	17	R
E	4	$13 \times 4 = 52$	$52 + 4 = 56$	4	E
R	17	$13 \times 17 = 221$	$221 + 4 = 225$	17	R

$K_1 = 13$ is not coprime with 26

The entire 26 letter alphabet is compressed into just two ciphertext letters (E & R)

7. Find the Inverse Matrix for the Key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$.

Ans. $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = (3 \times 7) - (5 \times 2) = 21 - 10 = 11$

$$11^{-1} \bmod 26 = 26 = 11(2) + 4 \quad \psi = 4 - 3(1) \\ 11 = 4(2) + 3 \quad \Rightarrow \quad 1 = 4 - 3(11 - 4(2)) \\ 1 = 4(2) - 3(11)$$

$$1 = (26 - 11(2)) - 3(11) = -6(11) - 3(11) + 3 \times 26$$

$$1 = (-7)11 + 3 \times 26, \quad -7 + 26 = 19$$

$$11^{-1} \bmod = 19$$

$$19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8. Find the inverse of the following matrix where entries are considered modulo 26:

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

Ans: $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \Rightarrow (11 \times 3) - (13 \times 2) = (33 - 26) = 7$

$$7^{-1} \text{ mod } 26 \Rightarrow 26 = 7(3) + 5 \Rightarrow 1 = 5 - 2(2)$$

$$7 = 5(1) + 2 \Rightarrow 1 = 5 - 2(7 - 5)$$

$$5 = 2(2) + 1 \Rightarrow 1 = 3(5) - 2(7)$$

$$\Rightarrow 1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-71)7 + 3(26)$$

$$-11 + 26 = 15$$

$$7^{-1} \text{ mod } 26 = 15$$

$$15 \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 165 & 195 \\ 30 & 45 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9. Is the Key valid? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

Ans: $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} = (2 \times 21) - (13 \times 14) = (42) - (182) = -140$

-140 is not coprime with 26

10. Encrypt using 2×2 Hill cipher for the plaintext "SOLVED" using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

Ans: For encryption: $C = P * K \text{ mod } 26$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 3 \times 11 \\ 7 \times 8 \end{bmatrix} = \begin{bmatrix} 33 \\ 56 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$$

23 is non-zero

23 is co-prime with 26

plaintext : SGLVED

$$\text{c.i)} P = \begin{bmatrix} S \\ 0 \end{bmatrix} = \begin{bmatrix} 18 \\ 14 \end{bmatrix}$$

$$\begin{matrix} K & P \\ \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} & \begin{bmatrix} 18 \\ 14 \end{bmatrix} \end{matrix} = \begin{bmatrix} 3 \times 18 + 8 \times 14 \\ 7 \times 18 + 11 \times 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} K \\ U \end{bmatrix}$$

$$\text{c.ii)} P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix}$$

$$\begin{matrix} K & P \\ \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} & \begin{bmatrix} 11 \\ 21 \end{bmatrix} \end{matrix} = \begin{bmatrix} 3 \times 11 + 8 \times 21 \\ 7 \times 11 + 11 \times 21 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{bmatrix} T \\ W \end{bmatrix}$$

$$\text{c.iii)} P = \begin{bmatrix} E \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{matrix} K & P \\ \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} & \begin{bmatrix} 4 \\ 3 \end{bmatrix} \end{matrix} = \begin{bmatrix} 3 \times 4 + 8 \times 3 \\ 7 \times 4 + 11 \times 3 \end{bmatrix} = \begin{bmatrix} 36 \\ 59 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 10 \\ 9 \end{bmatrix} = \begin{bmatrix} K \\ J \end{bmatrix}$$

Ciphertext : KUTWKJ